MILITARY SPECIFICATION

SIMULATOR, HAND CONTROL SM-214A/APQ-83

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be of one type and shall simulate the electrical input and output characteristics of the Aircraft Hand Control so that it shall be compatible with the Radar Set AN/APQ-83 or Radar Set AN/APQ-94.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-R-21642	Radar Set AN/APQ-83, Military Specification for
MIL-R-23373	Radar Set AN/APQ-94, Military Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-D-18300	Design Data Requirements for Contracts Covering Airborne Avionic Equipment
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

PUBLICATIONS

SAR-300	Material Changes and Material Bulletins, Preparation of

FSC 4920

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2.2 Availability of Documents

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

2.3 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, the following shall govern:

(1) Parts and materials as approved by Specification MIL-T-21200 shall be given first consideration.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-T-21200, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-T-21200.

3.2 Design and Construction

3.2.1 General - This equipment shall fulfill the applicable design, workmanship and construction requirements of Specification MIL-T-21200, Class 3, unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 3 equipment, except as modified herein.

3.2.2.1 Vibration - In accordance with Specification MIL-T-21200, paragraph 3.2.14.4, with these exceptions:

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Frequency Range	Double Amplitude
5 to 15	0.06 inch
15 to 25	0.04 inch
25 to 33	0.02 inch

A test at 0.02 inch double amplitude with the rate of frequency varying between 33 and 55 cycles per second is to be run for exploratory determination of mechanical transmissibility. This test is to be non-destructive.

3.2.2.2 Sand, Dust, Fungus and Explosion - Sand, dust, fungus and explosion requirements of Specification MIL-T-21200 shall not be applicable to this equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests.

Temperature	Room ambient ($30^{o} \pm 10^{o}C$)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity

3.2.4 Form Factor - The unit shall be housed in a rectangular metal case and shall not exceed 8 by 5-1/8 by 5-1/2 inches. The total weight of the equipment, excluding cables, shall not exceed 5 pounds. The case shall serve as a protective dust cover and shall not be subjected to drop tests or immersion tests.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without the necessity for readjustment of any controls.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 200 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.7 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.2.9 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.10 Maintenance Provision and Field Testing - Provision for maintenance shall be as specified in Specification MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

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3.2.11 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-T-21200.

3.2.12 Input Power - The input power shall be obtained from the radar set and shall consist of the following: +15 V dc at 1.5 ma maximum and -15 V dc at 1.5 ma maximum and +150 V dc at 10 ma maximum.

3.3 Detail Requirements

3.3.1 Simulator, Hand Control SM-214A/APQ-83 - This unit shall permit the operator to control the Radar Set in the following manner:

3.3.1.1 Search Functions

3.3.1.1.1 Azimuth - The azimuth control adjusts the position of the acquisition circle in azimuth to determine target azimuth before acquisition.

3.3.1.1.2 Tilt - The tilt control adjusts the tilt angle of the antenna search pattern and positions the tilt mark on the CRT display to indicate this tilt angle.

3.3.1.1.3 Range - The range control adjusts the range of the acquisition circle to the range of the target before acquisition.

3.3.1.1.4 Scan - Controls shall be included to enable those radar systems so equipped to operate in a right, center, or left narrow scan mode of operation.

3.3.1.2 Acquisition Functions - When the Action Button is held depressed, the Radar Set shall operate in the acquisition mode. In this mode, the Simulator, Hand Control, shall provide the following functions:

3.3.1.2.1 Azimuth - The azimuth control adjusts the antenna azimuth to manually designate the target azimuth as though using a searchlight beam.

3.3.1.2.2 Tilt - The tilt control adjusts the antenna tilt angle to manually designate the target elevation as though using a searchlight beam and also positions the tilt mark.

3.3.1.2.3 Range - The range control adjusts the range of the acquisition circle to coincide with the range of the target before lock-on.

3.3.1.3 Track Functions - After the Radar has locked-on the target, the Simulator, Hand Control, shall have no effect on the opreration, except that actuation of the Action Button shall return the Radar to the acquisition mode.

3.3.1.4 Function Range - The knob on the front panel shall permit operation of the following functions within the range specified.

3.3.1.4.1 Azimuth - The azimuth control range of the simulator shall be a minimum of left 44° to right 44°.

3.3.1.4.2 Tilt - The tilt control range of the simulator shall be a minimum of up 42° to down 27°.

3.3.1.4.3 Range - The range control shall permit adjustment of the minimum range strobe a minimum of from 600 yards to 35,000 yards.

3.3.1.5 Mounting - This unit shall be panel-mounted.

3.3.1.6 Connector - A connector shall be provided on the rear of the simulator to connect the simulator to the Test Bench Harness AN/APM-145.

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3.3.1.7 Finish - The case shall be finished with gray enamel conforming to MIL-T-21200. The panel shall be finished with black synthetic lusterless enamel in accordance with TT-E-527.

3.4 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified.

3.5 Preproduction Sample - Unless otherwise specified in the contract or order, a preproduction sample of the Simulator, Hand Control SM-214A/APQ-83 shall be manufactured such that it is representative of the production lot in all respects. The sample will be tested as specified in section 4 herein and is for the purpose of determining that the contractor,prior to his starting production is capable of yielding items that comply with the technical requirements of the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Tests Required - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests

(2) Sample Tests

(3) Acceptance Test

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the procuring activity for review and approval prior to shipping the equipment to the specified destination for service approval tests. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with MIL-T-5422 to the extent specified herein, interference tests in accordance with MIL-I-6181 and drop tests for transit and combination cases in accordance with Specification MIL-T-21200.

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4.2.3 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract.

4.3 Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and shall have successfully passed the individual tests.

4.3.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following test:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction and special tests.

4.3.2 Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Upon failure of the production sample to pass any of the tests, delivery of equipment under the contract shall cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Special Tests

(3) Life Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated at the conditions specified herein a total accrued running time of 6 hours. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a maximum double amplitude of 0.02 inch.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes as specified herein. The direction of vibration should be vertical to the normal mounting plane. The equipment shall be non-operating during the 10 minute vibration period. The 6-hour period of operation shall include that time necessary for performing all equipment tests and calibrations. Only that time necessary to make a total accrued running time of 6 hours is required for operation after the 10-minute vibration period. During the 6-hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test continued. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.2.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment.

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require variations to assure satisfactory control.

4.4.2.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minute cycling in each of 3 planes)

(3) Humidity Tests

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4.4.3 Life Tests - When requested by the procuring activity, one unit shall be selected from current production equipment and shall have successfully passed the Individual Tests.

4.4.3.1 Scope of Test - A 200 hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life test shall be performed under the conditions specified in 4.4.3.1.1. A performance check shall be made to determine the equipment is operating satisfactorily.

4.4.3.1.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ±2.0 volts

4.4.3.1.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3.1.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.4.3.2 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

4.4.3.3 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

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(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the procuring activity. To obtain procuring activity approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a procuring activity's recommended change or a contractor's proposed change. When required, the contractor shall also prepare a Material Change (MC) and a Material Bulletin (MB) to cover the approved change. Engineering Change Proposals, Material Changes and Material Bulletins shall be prepared in accordance with SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVWEPS FORM 4121/3 (8-62)

FORM APPROVED
BUDGET BUREAU NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by Government or contractor personnel involved in the use of this specification in procurement of products for ultimate use by the activity shown on the reverse of this sheet.

This sheet is provided for obtaining information which will aid the activity shown in insuring that through use of this specification suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this sheet will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to the activity shown.

SPECIFICATION *(No. and abbrev. title)*

ORGANIZATION	CITY	STATE

CONTRACT NUMBER	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THIS SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
(If so, give paragraph number and wording, and recommendations for correcting the deficiencies.)

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
☐ YES ☐ NO (IF YES, IN WHAT WAY?)

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If additional papers are attached, send this form and papers in an envelope. This form is addressed for use in window envelope when appropriate.)*

SUBMITTED BY (Print name and title)	DATE

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BUREAU OF NAVAL WEAPONS
WASHINGTON 25, D.C.

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